

Package ‘AnnotatoR’

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Title An R package for the creation of general purpose, IonReporter-compatible variant annotation files.

Version 1.0

Authors Dionysios Fanidis [aut, cre] <https://orcid.org/0000-0002-4053-2090>

Description Given a set of HGNC gene symbols, the package retrieves on demand known variants and respective annotation from publicly available databases/predictors. Although formatted as IonReporter-compatible files, results can be used for any other purpose.

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Encoding UTF-8

Imports EnsDb.Hsapiens.v86,
EnsDb.Hsapiens.v75,
BSgenome.Hsapiens.UCSC.hg38,
ensemblDb,
GenomicRanges,
HGNChelper,
httr,
jsonlite,
IRanges,
progressr,
rjson,
Rsamtools,
S4Vectors,
stringr,
utils,
SummarizedExperiment,
VariantAnnotation,
liftOver,
dplyr

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annotate	<i>Create general purpose, IonReporter-compatible variant annotation files.</i>
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Description

A wrapper function for downloading and organizing annotation files.

Usage

```

annotate(
  gns,
  annotators = c("revel", "alphamissense", "clinvar_sig", "gnomad_af", "intervar",
    "mutation_taster", "all"),
  databases = c("gnomad_man", "gnomad_auto", "clinvar", "lovd3", "all"),
  panelName = "./results/panel",
  path = "./dbs",
  type = c("exomes", "genomes", "both"),
  liftover = FALSE,
  saveRaw = FALSE
)

```

Arguments

gns	Gene names (HGNC) character vector.
annotators	Annotator resources to retrieve data from.
databases	Databases to retrieve variants from. The argument is taken into consideration when 'intervar', 'gnomad_af', 'mutation_taster' or 'all' annotators are selected. 'gnomad_man' expects manually downloaded gnomAD csv files placed in 'gnomad/manual' subdirectory of the directory set in 'path' argument (see below).
panelName	Path and prefix for output file.
path	Path to download directory.
type	Type of gnomAD data to download. Required when 'annotators' argument is set to/includes 'gnomad_af', 'mutation_taster', 'intervar', 'all'
liftover	Should variants be lifted over to hg19? (Default hg38)
saveRaw	Save all variants without any annotation. Applicable only for 'intervar' and 'all' annotators.

Value

VCF-like annotation files compatible with IonReporter.

annotateInterVar	<i>Annotate via InterVar's API.</i>
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Description

Predict variants' pathogenicity according to ACMG criteria as implemented by InterVar.

Usage

```
annotateInterVar(vcf, liftover)
```

Arguments

vcf	A variant table as the one created by 'collectVars'
liftover	Were variants lifted over?

Value

A nested list with InterVar's ACMG and other annotation pieces per variant.

clinvar_fetch	<i>Fetch ClinVar variants.</i>
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Description

Collect known variants for specific genes from ClinVar

Usage

```
clinvar_fetch(gns, path, liftover)
```

Arguments

gns	Gene names (HGNC) character vector.
path	Path to download directory.
liftover	Should variants be lifted over?

Value

A VCF-like data.frame with ClinVar variants.

collectVars	<i>Collect known variants.</i>
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Description

A wrapper function to collect known variants for specific genes from public resources (gnomAD, ClinVar, LOVD3).

Usage

```
collectVars(
  gns,
  databases = c("gnomad_man", "gnomad_auto", "clinvar", "lovd3", "all"),
  path,
  type = c("exomes", "genomes", "both"),
  liftover
)
```

Arguments

gns	Gene names (HGNC) character vector.
databases	Databases to retrieve variants from.
path	Path to download directory.
type	Type of gnomAD data to download. Used when 'gnomad_auto' database is selected.
liftover	Should variants be lifted over?

Value

A VCF-like data.frame with gene variants from the selected databases.

constructAF	<i>Retrieve gnomAD allele frequencies.</i>
-------------	--

Description

A wrapper function to retrieve population allele frequencies as recorded in gnomAD. 'gnomad_man' searches for manually downloaded gnomAD csv files in 'gnomad/manual' inside the 'dbs' directory. If not found, it downloads (upon approval) gnomAD raw files instead.

Usage

```
constructAF(
  gns,
  path,
  databases = c("gnomad_man", "gnomad_auto"),
  type = c("exomes", "genomes", "both"),
  liftover
)
```

Arguments

gns	Gene names (HGNC) character vector.
path	Path to download directory.
type	Type of gnomAD data to download. Used when 'gnomad_auto' database is selected.
liftover	Should variants be lifted over?

Value

A VCF-like data.frame with gnomAD allele frequency in the 3rd column ('ID').

constructAM	<i>Construct an AlphaMissense annotation file.</i>
-------------	--

Description

Download and organize AlphaMissense predictions for genes of interest.

Usage

```
constructAM(gns, path, liftover)
```

Arguments

<code>gns</code>	Gene names (HGNC) character vector.
<code>path</code>	Path to download directory.
<code>liftover</code>	Should variants be lifted over?

Value

A VCF-like data.frame with AlphaMissense pathogenicity prediction in the 3rd column ('ID').

<code>constructClinVar</code>	<i>Retrieve ClinVar significance.</i>
-------------------------------	---------------------------------------

Description

A wrapper function to retrieve ClinVar variant significance from gnomAD. Manually downloaded gnomAD files are required.

Usage

```
constructClinVar(gns, path, liftover)
```

Arguments

<code>gns</code>	Gene names (HGNC) character vector.
<code>path</code>	Path to download directory.
<code>liftover</code>	Should variants be lifted over?

Value

A VCF-like data.frame with ClinVar germline classification values in the 3rd column ('ID').

<code>constructClinVar_deprecated</code>	<i>Construct ClinVar annotation file.</i>
--	---

Description

Download and organize ClinVar significance classification with review status level for specific genes.

Usage

```
constructClinVar_deprecated(gns, path, liftover)
```

Arguments

gns	Gene names (HGNC) character vector.
path	Path to download directory.
liftover	Should variants be lifted over?

Value

A VCF-like data.frame with ClinVar germline classification values and number of review stars in the 3rd column ('ID').

constructREVEL	<i>Construct REVEL annotation file.</i>
----------------	---

Description

Download and organize REVEL meta-predictions for specific genes.

Usage

```
constructREVEL(gns, path, liftover)
```

Arguments

gns	Gene names (HGNC) character vector.
path	Path to download directory.
liftover	Should variants be lifted over?

Value

A VCF-like data.frame with REVEL deleteriousness prediction score in the 3rd column ('ID').

download_am	<i>Download AlphaMissense tsv files</i>
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Description

This function downloads AlphaMissense tsv file.

Usage

```
download_am(path)
```

Arguments

path	Where to download data
------	------------------------

Value

AlphaMissense vcf files

download_clinvar	<i>Download ClinVar files</i>
------------------	-------------------------------

Description

This function downloads ClinVar vcf file.

Usage

```
download_clinvar(path)
```

Arguments

path Where to download data

Value

ClinVar vcf file.

download_gnomad	<i>Global functions Download gnomad chromosome files</i>
-----------------	--

Description

Downloads gnomAD vcf files according to specified chromosome.

Usage

```
download_gnomad(chr, type = c("exomes", "genomes", "both"), path)
```

Arguments

chr Chromosome name
type Type of gnomAD data to download.
path Where to download data

Value

gnomAD vcf files

download_lovd3	<i>Download LOVD3 gene variants.</i>
----------------	--------------------------------------

Description

This function downloads LOVD3 gene.

Usage

```
download_lovd3(gns, path)
```

Arguments

gns	Gene names
path	Where to download data

Value

LOVD3 gene files.

download_revel	<i>Download REVEL chromosome files.</i>
----------------	---

Description

This function downloads REVEL vcf files per chromosome.

Usage

```
download_revel(chr, path)
```

Arguments

chr	Chromosome name
path	Where to download data

Value

REVEL vcf files.

gnomad_fetch_auto	<i>Fetch gnomAD variants.</i>
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Description

Collect known variants for specific genes from gnomAD.

Usage

```
gnomad_fetch_auto(  
  map,  
  type = c("exomes", "genomes", "both"),  
  path,  
  liftover,  
  af = FALSE  
)
```

Arguments

map	Gene data as in map.
type	Type of gnomAD data to download. Used when 'gnomad_auto' database is selected.
path	Path to download directory.
liftover	Should variants be lifted over?
af	Maintain allele frequencies?

Value

A VCF-like data.frame with gnomAD variants.

gnomad_fetch_manual	<i>Fetch manually downloaded gnomAD variants.</i>
---------------------	---

Description

Collect known variants for specific genes downloaded manually from gnomAD

Usage

```
gnomad_fetch_manual(map, path, liftover, af = FALSE, sig = FALSE)
```

Arguments

map	Gene data data.frame as returned from ‘map_fetch‘.
path	Path to download directory.
liftover	Should variants be lifted over?
af	Maintain allele frequencies?

Value

A VCF-like data.frame with gnomAD allele frequencies (af = TRUE) or ClinVar significance values (sig = TRUE) in the 3rd column (‘ID‘).

lift	<i>LiftOver hg38 to hg19 genomic coordinates using rtracklayer functionalities</i>
------	--

Description

LiftOver hg38 to hg19 genomic coordinates using rtracklayer functionalities

Usage

```
lift(vcf, ...)
```

Arguments

vcf	A vcf dataframe with CHR and POS columns
...	Arguments for methods

Value

A GRanges object of lifted over genomic coordinates

lovd3_fetch	<i>Fetch LOVD3 variants.</i>
-------------	------------------------------

Description

Collect known variants for specific genes from LOVD3

Usage

```
lovd3_fetch(gns, path, liftover)
```

Arguments

gns Gene names (HGNC) character vector.
path Path to download directory.
liftover Should variants be lifted over?

Value

A VCF-like data.frame with LOVD3 variants.

map_fetch	<i>Annotate genes using Ensembl data</i>
-----------	--

Description

Annotate genes using Ensembl data

Usage

```
map_fetch(ensdb, gns, trans = FALSE)
```

Arguments

ensdb An EnsDb package.
gns Gene names (HGNC) character vector.
trans Should transcript level data be returned?

Value

A data frame with genes (default) or ensembl transcripts as rows and Ensembl annotation as columns.

process_json	<i>Handling errors</i>
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Description

Handling errors

Usage

```
process_json(j)
```

Arguments

j json file

Value

LOVD3 gene files.

revstatus_map	<i>Review status on aggregate records (VCV and RCV) - ClinVar</i>
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Description

Review status on aggregate records (VCV and RCV) - ClinVar

Usage

revstatus_map

taste	<i>Annotate via MutationTaster's API.</i>
-------	---

Description

Retrieve variants' pathogenicity predictions from MutationTaster.

Usage

taste(vcf, liftover)

Arguments

- vcf A variant table as the one created by 'collectVars'.
- liftover Whether variants have been lifted over (boolean).

Value

Annotated variants with MutationTaster2021 and MutationTaster2025 predictions for GRCh37 (liftover = TRUE) and GRCh38 (liftover = FALSE) variants, respectively.

vcf_header_allele	<i>VCF headers for IonReporter</i>
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Description

VCF headers for IonReporter

Usage

vcf_header_allele

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